

COMPUTER NETWORKS

Complete Notes — Basic to Advance

Hinglish Medium | Interview Prep | Diagrams & Examples



Topics Covered: OSI Model • TCP/IP • Subnetting • Routing • Security • DNS • HTTP • TCP • UDP • VPN • CDN • Real-World Flows



Purpose: Exam + Interview Preparation — Sabhi topics ko clearly samajhne ke liye

CHAPTER 1: INTRODUCTION TO COMPUTER NETWORKS

1.1 What is a Computer Network?

Definition

Computer Network ek aisa system hai jisme do ya zyada computers ko ek saath connect kiya jaata hai taaki woh data, resources aur information share kar sakein.

Simple words mein: 'Computers ka ek group jo ek doosre se baat kar sakein.'

Working (Step-by-Step)

1. Sender computer data ko packets mein divide karta hai.
2. Packets network ke through travel karte hain (wires ya wireless).
3. Intermediate devices (routers, switches) packets ko sahi jagah pahunchate hain.
4. Receiver packets ko receive karke data ko reassemble karta hai.

Diagram

```
[Computer A] — [Switch] — [Router] — [Internet] — [Computer B]
```

Real-Life Example

 **Example:** Jab aap Google.com open karte ho — aapka computer Google ke server se network ke through connect hota hai aur webpage download karta hai.

Why & How it Works

- Why: Resource sharing (printer, files, internet), communication (email, chat), centralized data storage.
- How: Electrical signals (wired) ya radio waves (wireless) ke through data travel karta hai, protocols ensure karte hain ki data sahi format mein jaaye.

Interview Questions

-  **Q1:** Computer Network kya hota hai? Uske main advantages kya hain?
-  **Q2:** Network ke bina modern internet possible hota? Kyon?

1.2 Types of Networks — LAN, MAN, WAN

Type	Full Form	Range	Speed	Real-Life Example
LAN	Local Area Network	Room / Building	100 Mbps–10 Gbps	School lab, Office
MAN	Metropolitan Area Network	City	10–100 Mbps	City-wide cable TV
WAN	Wide Area Network	Countries / Globe	Varies	Internet, Bank ATM network

💡 **Yaad rakhne ki trick:** LAN = Local (locality mein), MAN = Metro (city mein), WAN = Wide (poori duniya mein)

🔗 Interview Questions

? **Q1:** LAN aur WAN mein kya difference hai? Example do.

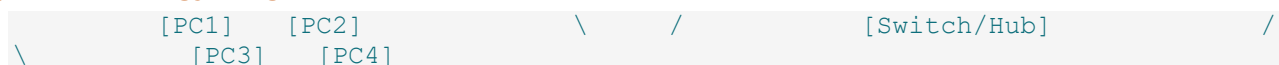
? **Q2:** MAN ka real-life use case kya hai?

1.3 Network Topologies

Topology matlab network mein computers/devices ka physical ya logical arrangement.

Topology	Structure	Advantage	Disadvantage
Bus	Sab ek single cable se	Cheap, easy setup	Ek failure se poora band
Star	Sab ek central hub/switch se	Easy to add/remove, fault isolation	Hub fail = poora network fail
Ring	Devices circle mein connected	Equal access, no collision	Ek failure = ring break
Mesh	Har device baaki sabse connected	High redundancy, fault tolerant	Expensive, complex wiring

📖 Star Topology Diagram (Most Common)



💡 **Real Example:** Ghar mein WiFi router se sab phones connect hote hain = Star Topology

? **Interview Q:** Star vs Mesh topology mein kya difference hai? Kab kaunsa use karein?

1.4 Network Components

Device	Kya karta hai	OSI Layer	Key Point
Hub	Sabko data broadcast karta hai	Layer 1 (Physical)	Dumb device, no filtering
Switch	Specific device ko data bhejta hai	Layer 2 (Data Link)	MAC address use karta hai
Router	Networks ke beech route karta hai	Layer 3 (Network)	IP address use karta hai
Bridge	Do LAN segments join karta hai	Layer 2 (Data Link)	Traffic filter karta hai
Modem	Digital ↔ Analog convert	Layer 1 (Physical)	ISP se connect hota hai

 **Hub vs Switch:** Hub sabko data bhejta hai (broadcast), Switch sirf sahi device ko data bhejta hai (unicast). Switch smarter hai.

 **Interview Q:** Router aur Switch mein kya difference hai? Dono kaon se layer pe kaam karte hain?

1.5 Performance Metrics

Bandwidth

Bandwidth = Network ki maximum data transfer capacity per second. Unit: bps, Kbps, Mbps, Gbps.

 **Analogy:** Bandwidth = Road ki width. Zyada wide road = zyada cars (data) ek saath ja sakti hain.

Latency

Latency = Data ko source se destination tak pahunchne mein lagaa hua time. Unit: milliseconds (ms).

 **Example:** Ping command se latency check karte hain: 'ping google.com' → Reply time = latency

Throughput

Throughput = Actual mein kitna data successfully transfer hua. Bandwidth se hamesha kam hota hai.

 **Formula:** Throughput ≤ Bandwidth. Real-world mein: interference, congestion, errors = throughput kam hota hai.

Metric	Definition	Unit	Example
Bandwidth	Max capacity	Mbps/Gbps	100 Mbps broadband
Latency	Travel time	ms	20ms ping to Google
Throughput	Actual speed	Mbps	80 Mbps actual download

 **Interview Q:** Bandwidth aur throughput mein kya difference hai? Ek example do.

CHAPTER 2: OSI MODEL (CORE)

2.1 Introduction to OSI Model

OSI = Open Systems Interconnection. Ye ek conceptual framework hai jo network communication ko 7 layers mein divide karta hai.



Why OSI?: Different companies ke devices ka ek saath kaam karna possible banane ke liye. ISO ne 1984 mein banaya. Standardization ke liye.

2.2 All 7 Layers — Overview

Layer #	Layer Name	Data Unit	Main Function	Key Protocols
7	Application	Data	User interaction	HTTP, FTP, DNS, SMTP
6	Presentation	Data	Encryption, Compression	SSL/TLS, JPEG, ASCII
5	Session	Data	Session management	NetBIOS, RPC
4	Transport	Segment	End-to-end delivery	TCP, UDP
3	Network	Packet	Routing & IP	IP, ICMP, ARP
2	Data Link	Frame	MAC, Error detection	Ethernet, Wi-Fi, PPP
1	Physical	Bits	Physical transmission	Cables, Hubs, NIC



Trick to Remember: All People Seem To Need Data Processing → A(7) P(6) S(5) T(4) N(3) D(2) P(1)

2.3 Layer-wise Deep Study

2.3.1 Physical Layer (Layer 1)

Definition

Physical layer bits ko electrical signals, light ya radio waves mein convert karta hai aur physical medium pe transmit karta hai.

Working

5. Data bits (0s and 1s) ko encoding karta hai.
6. Transmission medium ke through bits travel karte hain.

7. Receiver pe wapas decode hote hain.

Diagram

```
[PC: 0101...] —[NIC]—[Cable/Wi-Fi]—[NIC]—[PC: 0101...]
```

Real Example

 **Example:** Ethernet cable mein electrical signal, Fiber optic mein light pulses, WiFi mein radio waves.

 **Interview Q:** Physical layer ka kaam kya hai? Hub kaunse layer pe kaam karta hai?

2.3.2 Data Link Layer (Layer 2)

Definition

Data Link Layer frames banata hai, MAC addressing handle karta hai, aur error detection karta hai same network ke devices ke beech.

Working

8. Network layer se packet milti hai.
9. Frame banata hai: Header + Data + Trailer.
10. Source aur Destination MAC address add karta hai.
11. CRC (Cyclic Redundancy Check) se error detect karta hai.

Frame Structure

```
| Preamble | Dest MAC | Src MAC | Type | DATA | CRC |
```

 **MAC Address:** 48-bit address, hardware mein hardcoded hota hai. Format: AA:BB:CC:DD:EE:FF. Unique per device.

 **Interview Q:** MAC address aur IP address mein kya difference hai?

2.3.3 Network Layer (Layer 3)

Definition

Network Layer routing handle karta hai — packets ko source se destination tak best path se pohanchata hai. IP addressing yahan hoti hai.

Working

12. Data Link se frame milti hai, IP packet extract hoti hai.
13. Destination IP check karta hai.
14. Routing table se best path dhundta hai.
15. Packet next hop pe forward karta hai.

Routing Diagram

```
[PC 192.168.1.1] → [Router A] → [Router B] → [Server 8.8.8.8]  
↑ routing table ↑
```

? Interview Q: Router kaunse layer pe kaam karta hai? Routing kaise hoti hai?

2.3.4 Transport Layer (Layer 4)

Definition

Transport Layer end-to-end communication ensure karta hai. Data ko segments mein todta hai aur reliable delivery provide karta hai (TCP) ya fast delivery (UDP).

Feature	TCP	UDP
Connection	Connection-oriented (3-way handshake)	Connectionless
Reliability	Guaranteed delivery	No guarantee
Speed	Slower	Faster
Use Case	HTTP, FTP, Email	Video streaming, DNS, Gaming
Error Check	Yes (ACK)	No (best effort)

? Interview Q: TCP aur UDP mein kya difference hai? WhatsApp video call mein kaunsa use hoga?

2.3.5 Session Layer (Layer 5)

Session layer connections ko establish, manage, aur terminate karta hai. Multiple sessions ko track karta hai.

 **Real Example:** Jab aap ek hi time pe Facebook aur Gmail dono open karte ho — Session layer dono sessions ko alag-alag manage karta hai.

2.3.6 Presentation Layer (Layer 6)

Presentation layer data ko ek format se doosre format mein convert karta hai, encryption/decryption karta hai, aur compression handle karta hai.

- Encoding: ASCII, Unicode
- Compression: ZIP, JPEG, MP3
- Encryption: SSL/TLS

? Interview Q: HTTPS mein SSL/TLS kaunse layer pe kaam karta hai?

2.3.7 Application Layer (Layer 7)

Application layer end-user ke saath directly interact karta hai. Network services users ko provide karta hai.

 **Protocols:** HTTP (web), HTTPS (secure web), FTP (file transfer), DNS (domain resolution), SMTP (email), DHCP (IP allocation).

2.4 Encapsulation & Decapsulation

Definition

Encapsulation = Data bhejte waqt har layer apna header (aur sometimes trailer) add karta hai.

Decapsulation = Receive karte waqt har layer apna header remove karta hai.

Step-by-Step Data Flow

Layer	Action (Sending)	Data Unit	Added Info
7 - Application	Data create karta hai	Data	Application data
6 - Presentation	Encrypt/compress karta hai	Data	Format info
5 - Session	Session ID add karta hai	Data	Session header
4 - Transport	Segment banata hai	Segment	Port numbers
3 - Network	Packet banata hai	Packet	IP addresses
2 - Data Link	Frame banata hai	Frame	MAC addresses + CRC
1 - Physical	Bits transmit karta hai	Bits	0s and 1s

 **Yaad rakhne ki trick:** Sender: Data → Segment → Packet → Frame → Bits (top to bottom mein header add hota hai). Receiver: Bilkul ulta!

 **Interview Q:** Encapsulation aur Decapsulation kya hai? Step by step explain karo.

CHAPTER 3: TCP/IP MODEL

3.1 Introduction to TCP/IP

TCP/IP model real-world internet ka actual working model hai. OSI model theoretical hai, TCP/IP practical. Internet aaj TCP/IP pe hi chalta hai.



History: DARPA ne 1970s mein banaya. OSI se pehle aya. 4 layers hain (OSI ke 7 ke jagah).

3.2 OSI vs TCP/IP Mapping

OSI Layer	TCP/IP Layer	Protocols
Application (7)	Application	HTTP, FTP, DNS, SMTP, DHCP
Presentation (6)	Application	SSL/TLS, MIME
Session (5)	Application	NetBIOS
Transport (4)	Transport	TCP, UDP
Network (3)	Internet	IP, ICMP, ARP
Data Link (2)	Network Access	Ethernet, Wi-Fi
Physical (1)	Network Access	Cables, NIC

3.3 TCP/IP Layers Deep Dive

Network Access Layer

OSI ke Physical + Data Link ko combine karta hai. Hardware level pe kaam karta hai — NIC, cables, MAC addresses.

Internet Layer

IP addressing aur routing handle karta hai. Main protocols: IP, ICMP, ARP.



ICMP: Internet Control Message Protocol — ping aur traceroute yahi use karta hai. Error reporting ke liye.

Transport Layer

TCP (reliable) aur UDP (fast) — same as OSI Transport layer.

Application Layer

OSI ke Application + Presentation + Session teen layers ko ek mein combine karta hai. User-facing protocols yahan hain.



Interview Q: OSI aur TCP/IP mein kya fundamental differences hain?

CHAPTER 5: NETWORK LAYER (IP, ROUTING)

5.1 IP Addressing (IPv4)

Definition

IP Address ek 32-bit unique address hai jo har device ko internet/network pe identify karta hai.

Structure

192 . 168 . 1 . 1 [8 bits].[8 bits].[8 bits].[8 bits] = 32 bits
total

Class	Range	Default Mask	Use
A	1.0.0.0 – 126.255.255.255	255.0.0.0 (/8)	Large networks (ISPs)
B	128.0.0.0 – 191.255.255.255	255.255.0.0 (/16)	Medium networks
C	192.0.0.0 – 223.255.255.255	255.255.255.0 (/24)	Small networks (homes)
D	224.0.0.0 – 239.255.255.255	N/A	Multicast
E	240.0.0.0 – 255.255.255.255	N/A	Reserved/Research

 **Private IP Ranges:** 10.x.x.x (Class A), 172.16–31.x.x (Class B), 192.168.x.x (Class C) — ye internet pe route nahi hote.

5.2 Subnetting (VERY IMPORTANT)

Definition

Subnetting = Ek bada network ko chhote parts (subnets) mein todna. CIDR notation use hoti hai.

CIDR Notation

 **Format:** 192.168.1.0/24 → /24 matlab 24 bits network ke liye reserved, baaki 8 bits hosts ke liye.

CIDR	Subnet Mask	Hosts Available	Example
/24	255.255.255.0	254 hosts	192.168.1.0/24
/25	255.255.255.128	126 hosts	192.168.1.0/25
/26	255.255.255.192	62 hosts	192.168.1.0/26
/28	255.255.255.240	14 hosts	192.168.1.0/28
/30	255.255.255.252	2 hosts	192.168.1.0/30 (P2P links)

Subnetting Practice Example

Question: 192.168.10.0/26 ka Network Address, Broadcast, aur usable hosts kya honge?

```
/26 = 255.255.255.192    Network:    192.168.10.0    Broadcast: 192.168.10.63
Hosts:      192.168.10.1 to 192.168.10.62 (62 hosts)    Formula:    2^(32-26) - 2 =
2^6 - 2 = 62 hosts
```

? Interview Q: 192.168.1.0/28 network mein kitne usable hosts honge? Network aur broadcast address kya hai?

5.3 IPv6 Basics

IPv6 = 128-bit address. IPv4 addresses khatam ho rahe the, isliye IPv6 banaya gaya.

💡 Format: 2001:0db8:85a3:0000:0000:8a2e:0370:7334 → Colon-separated hexadecimal. Total: 3.4×10^{38} addresses!

? Interview Q: IPv4 aur IPv6 mein kya differences hain? IPv6 kyun zaroori hai?

5.4 Routing

Type	Definition	Advantage	Disadvantage
Static Routing	Admin manually routes set karta hai	Simple, secure	Scalable nahi
Dynamic Routing	Routers khud routes seekhte hain	Scalable, adaptive	Complex, overhead

5.5 Routing Algorithms

Distance Vector (RIP)

Har router apne neighbors ko apni routing table share karta hai. Bellman-Ford algorithm use karta hai. Simple par slow convergence.

Link State (OSPF)

Har router poore network ka complete map banata hai. Dijkstra algorithm use karta hai. Fast convergence, complex.

? Interview Q: Distance Vector aur Link State routing mein kya difference hai?

CHAPTER 6: TRANSPORT LAYER

6.1 TCP vs UDP

Feature	TCP	UDP
Full Form	Transmission Control Protocol	User Datagram Protocol
Connection	Connection-oriented	Connectionless
Reliability	Guaranteed (ACK)	No guarantee
Order	In-order delivery	May be out of order
Speed	Slower (overhead)	Faster
Header Size	20 bytes	8 bytes
Use Cases	HTTP, HTTPS, FTP, Email	DNS, Video, VoIP, Gaming

6.2 TCP Deep Dive

3-Way Handshake (Connection Establishment)

TCP connection establish karne ke liye 3 steps hote hain:

```
Client          Server          |          |          |---- SYN (seq=100)
----->| Step 1: Client bolta hai 'connect karo' |          |          |
|<--- SYN-ACK (seq=200, | Step 2: Server accept karta hai |          | ack=101) ----
-----|          |          |          |---- ACK (ack=201) ----->|
Step 3: Client confirm karta hai |          |          | === CONNECTION
OPEN === |
```

 **SYN-ACK:** SYN = Synchronize (connection start), ACK = Acknowledgment (confirm karna). 3-way isliye kyunki dono sides ko sync hona hai.

4-Way Termination (Connection Close)

```
Client          Server          |          |          |---- FIN -----
----->| Step 1: Client bolta hai 'khatam karo' |<--- ACK -----|
Step 2: Server ACK bhejta hai |<--- FIN -----| Step 3: Server apna FIN
bhejta hai |---- ACK ----->| Step 4: Client ACK bhejta hai | ===
CONNECTION CLOSED = |
```

Flow Control

Sender itna data na bheje ki receiver handle na kar sake. TCP sliding window use karta hai.

 **Sliding Window:** Receiver apni window size batata hai (kitna data ek baar receive kar sakta hai). Sender uss limit se zyada nahi bhejta.

Congestion Control

Network overload hone se bachana. Algorithms: Slow Start, Congestion Avoidance, Fast Retransmit.

 **Interview Q:** TCP ka 3-way handshake explain karo. Flow control kya hai?

6.3 UDP

UDP lightweight protocol hai. No handshake, no acknowledgment. Fast delivery chahiye tab use karo.

- DNS queries: Fast resolution chahiye
- Video streaming (YouTube): Kuch frames drop okay, lag nahi chahiye
- Online gaming: Real-time response priority
- VoIP calls: Low latency priority

? Interview Q: YouTube video stream UDP use karta hai ya TCP? Kyun? Agar packet drop ho toh kya hoga?

CHAPTER 7: APPLICATION LAYER PROTOCOLS

7.1 HTTP / HTTPS

Definition

HTTP = HyperText Transfer Protocol. Web browsers aur servers ke beech communication ka protocol.

Request-Response Cycle

```
Browser                                Web Server                                |
|-- GET /index.html HTTP/1.1 ----->| Client Request          | Host: www.example.com
|                                     |<-- HTTP/1.1 200 OK -----|
Server Response          | Content-Type: text/html          | <html>...</html>
```

HTTP Method	Use	Example
GET	Data fetch karna	Web page load karna
POST	Data send karna	Form submit, login
PUT	Data update karna	Profile update
DELETE	Data delete karna	Post delete karna

 **HTTPS:** HTTP + SSL/TLS = HTTPS. Data encrypted hota hai. Port 443 use karta hai. HTTP port 80.

 **Interview Q:** HTTP aur HTTPS mein kya difference hai? GET vs POST kab use karte hain?

7.2 DNS — Domain Name System

Definition

DNS ek phone book ki tarah hai internet ke liye. Domain name (google.com) ko IP address (142.250.80.46) mein convert karta hai.

DNS Resolution Steps

```
Browser → Local Cache → OS Cache → ISP DNS Resolver → Root DNS Server → TLD Server
(.com)      → Authoritative DNS Server → IP Address Return
```

 **DNS Port:** DNS UDP port 53 use karta hai (quick queries ke liye). Large responses ke liye TCP 53.

 **Interview Q:** Jab aap google.com type karte ho toh DNS kaise kaam karta hai? Step by step batao.

7.3 FTP (File Transfer Protocol)

FTP files ko network pe transfer karne ke liye use hota hai. Port 21 (control), Port 20 (data). Clear text (insecure). SFTP = FTP + SSH (secure).

7.4 SMTP (Simple Mail Transfer Protocol)

Email bhejne ke liye use hota hai. Port 25 (server-to-server), Port 587 (client-to-server). Gmail, Outlook sab SMTP use karte hain.

Sender → SMTP → Mail Server → Receiver Mail Server → POP3/IMAP → Inbox

7.5 DHCP (Dynamic Host Configuration Protocol)

Definition

DHCP automatically IP addresses assign karta hai devices ko jab woh network se connect hote hain.

DORA Process

D - Discover: Client broadcasts 'Mujhe IP chahiye' O - Offer: DHCP server offers IP address R - Request: Client accepts the offer A - ACK: Server confirms IP assignment

 **Example:** Ghar mein WiFi se connect hote ho toh DHCP hi aapko automatically 192.168.1.x address deta hai.

 **Interview Q:** DHCP ka DORA process explain karo. Static IP aur Dynamic IP mein kya difference hai?

CHAPTER 8: NETWORK SECURITY

8.1 Basics of Security

Network security ka goal hai: Unauthorized access rokna, data protect karna, aur availability ensure karna.

CIA Triad	Definition	Example
Confidentiality	Data sirf authorized log dekh sakein	Encryption
Integrity	Data tamper na hua ho	Hash (SHA-256)
Availability	Service hamesha available ho	Load balancer, backups

8.2 Encryption

Type	Definition	Key	Example
Symmetric	Same key for encrypt + decrypt	1 key shared	AES, DES
Asymmetric	Public key encrypt, private decrypt	2 keys (public + private)	RSA, HTTPS

 **Analogy:** Symmetric = ek taala ek chaabi. Asymmetric = taala publicly available (anyone can lock), chaabi sirf aapke paas (only you can open).

8.3 SSL/TLS & HTTPS

How HTTPS works

```
1. Browser: 'Hello Server, ye ciphers support karta hoon' 2. Server: 'Ye cipher use karo, mera SSL certificate lo' 3. Browser: Certificate verify karta hai (CA check) 4. Session key exchange (asymmetric) 5. Data transfer (symmetric encryption with session key)
```

 **Interview Q:** HTTPS kaise kaam karta hai? SSL handshake explain karo.

8.4 Firewall

Firewall network traffic monitor karta hai aur rules ke basis pe allow/block karta hai.

Type	Working	Example
Packet Filter	IP/Port ke basis pe filter	Block port 23 (Telnet)
Stateful	Connection state track karta hai	Established connections allow
Application (WAF)	Layer 7 pe inspect karta hai	SQL injection block

8.5 VPN (Virtual Private Network)

VPN ek encrypted tunnel banata hai aapke device aur VPN server ke beech. Aapki real IP hide hoti hai.

[Your PC] ==== [Encrypted Tunnel] ==== [VPN Server] -----> [Internet]
(Private, Secure) (Public IP used)

8.6 Common Attacks

Attack	Definition	Prevention
DDoS	Bahut saare requests se server overwhelm karna	Rate limiting, CDN, Firewall
MITM	Attacker beech mein data intercept karta hai	HTTPS, Certificate Pinning
Phishing	Fake website/email se credentials churaana	User awareness, 2FA
SQL Injection	Database mein malicious SQL inject karna	Parameterized queries

? Interview Q: DDoS attack kya hai? Use kaise prevent karte hain?

CHAPTER 10: REAL-WORLD NETWORK FLOWS

10.1 What Happens When You Open a Website?

Ye sabse common interview question hai! Poori flow yahan hai:

Complete Flow: google.com open karna

```
STEP 1: URL Type karo | Browser checks local cache for google.com IP | Not found → OS checks hosts file | Not found → DNS Query | STEP 2: DNS Resolution | ISP DNS Resolver query karta hai | Root DNS → .com TLD → google.com Authoritative | Returns IP: 142.250.80.46 | STEP 3: TCP 3-Way Handshake | SYN → SYN-ACK → ACK (port 443 for HTTPS) | Connection established | STEP 4: TLS Handshake (for HTTPS) | Certificate exchange | Session key established | STEP 5: HTTP GET Request | GET / HTTP/1.1 | Host: www.google.com | STEP 6: Server Response | HTTP/1.1 200 OK | HTML, CSS, JS files return | STEP 7: Browser Render | HTML parse → DOM build → Page display!
```

? Interview Q: Jab aap browser mein URL type karte ho toh kya hota hai? DNS se lekar page display tak explain karo.

10.2 YouTube Video Streaming

Buffering aur CDN kaise kaam karta hai

16. Aap YouTube pe video click karte ho.
17. DNS resolve karta hai youtube.com ka IP.
18. YouTube CDN (Content Delivery Network) nearest server se video serve karta hai.
19. HTTP/HTTPS pe video chunks download hote hain (adaptive bitrate streaming).
20. Buffer full hone pe playback shuru hota hai.
21. Network slow ho toh quality automatically reduce hoti hai (360p, 480p, 1080p).

💡 CDN: Google ke CDN servers duniya bhar mein hain. Aapko nearest server se data milta hai = low latency.

💡 UDP?: YouTube QUIC protocol use karta hai (UDP based) for faster streaming. Kuch packet loss acceptable hai video mein.

10.3 WhatsApp Messaging — Packet Flow

22. Message type karo, Send dabao.
23. App message ko encrypt karta hai (End-to-End Encryption using Signal Protocol).
24. TCP connection to WhatsApp server (reliable delivery chahiye).
25. Server message ko receiver ko forward karta hai.
26. Receiver ka app decrypt karke message show karta hai.

🔒 E2E Encryption: WhatsApp mein sirf sender aur receiver message padh sakte hain — WhatsApp ke server bhi nahi. Asymmetric encryption use hoti hai.

? Interview Q: WhatsApp message ka packet flow explain karo. E2E encryption kya hai?

CHAPTER 11: ADVANCED & INTERVIEW TOPICS

11.1 NAT (Network Address Translation)

Definition

NAT private IP addresses ko public IP address mein convert karta hai jab packets internet pe jaate hain.

```
Home Network:      Internet:  192.168.1.2  ↘
192.168.1.3  ──┬── [Router/NAT] ── 203.0.113.1 ──> Google  192.168.1.4  ─┬── (one public
IP)          (one IP for all)
```

💡 **Why NAT?:** IPv4 addresses limited hain. NAT se hazaron devices ek public IP share kar sakte hain.

❓ **Interview Q:** NAT kya hai aur kyun zaroori hai? IPv6 aane se NAT ki zaroorat khatam ho jaayegi?

11.2 Load Balancer

Load balancer incoming traffic ko multiple servers mein distribute karta hai taaki koi ek server overloaded na ho.

```
└─> Server 3      ┌─> Server 1  Users ──> [Load Balancer] ─> Server 2
```

Algorithm	How it works
Round Robin	Ek ek karke har server ko request bhejta hai
Least Connections	Sabse kam connections wale server ko bhejta hai
IP Hash	Client IP se server decide hota hai (sticky sessions)

❓ **Interview Q:** Load balancer kya hai? Round robin algorithm explain karo.

11.3 Proxy vs Reverse Proxy

Type	Who uses it	Purpose	Example
Forward Proxy	Client side	Client ki identity hide karna	Corporate proxy, VPN
Reverse Proxy	Server side	Servers protect karna, load balance	Nginx, Cloudflare

```
Forward:  Client ─> [Proxy] ─> Internet ─> Server  Reverse:  Client ─> Internet ─> [Reverse
Proxy] ─> Server
```

❓ **Interview Q:** Proxy aur Reverse Proxy mein kya difference hai? Nginx kaisa use hota hai?

11.4 CDN (Content Delivery Network)

CDN = Globally distributed servers jo static content (images, videos, CSS, JS) cache karke users ke nearest server se serve karte hain.

- Latency reduce hoti hai (nearest server se serve hota hai)
- Origin server ka load reduce hota hai
- DDoS protection provide karta hai
- Examples: Cloudflare, AWS CloudFront, Akamai

 **Interview Q:** CDN kya hai? Isko use karne ke kya fayde hain?

11.5 Socket Programming Basics

Socket = IP Address + Port Number. Network pe do processes ke beech communication ka endpoint.

Server Side:	Client Side:
<code>connect()</code>	<code>socket() → bind() → listen() → socket() →</code>
<code>→ accept() → recv/send()</code>	<code>→ send/recv() → close()</code>
<code>→ close()</code>	

 **Port Numbers:** Well-known ports: HTTP=80, HTTPS=443, FTP=21, SSH=22, DNS=53, SMTP=25, DHCP=67/68

 **Interview Q:** Socket kya hai? Server-client model mein socket kaise kaam karta hai?

CHAPTER 12: NETWORK TOOLS

Tool	Command	Use	Example Output
ping	ping google.com	Latency check, reachability test	Reply from 142.250.x.x: time=20ms
tracert	tracert google.com	Packet ka route trace karna	Hop by hop path show karta hai
netstat	netstat -an	Active connections dekhna	TCP connections list
nslookup	nslookup google.com	DNS resolution check karna	Returns IP of domain
ipconfig/ifconfig	ipconfig (Win) / ifconfig	IP address dekhna	192.168.1.5 / Subnet mask
curl	curl -I https://example.com	HTTP headers dekhna	HTTP/1.1 200 OK headers

 **Ping vs Traceroute:** Ping = just check karo server alive hai ya nahi. Traceroute = poora path kaise ja raha hai woh dekho (kaun se routers se guzar raha hai).

 **Interview Q:** ping aur traceroute mein kya difference hai? ICMP kaunsa use karta hai?

QUICK REFERENCE — PORTS & PROTOCOLS

Protocol	Port	Transport	Purpose
HTTP	80	TCP	Web browsing (insecure)
HTTPS	443	TCP	Secure web browsing
FTP	20/21	TCP	File transfer
SSH	22	TCP	Secure remote access
Telnet	23	TCP	Remote access (insecure)
SMTP	25/587	TCP	Email sending
DNS	53	UDP/TCP	Domain name resolution
DHCP	67/68	UDP	IP auto-assignment
POP3	110	TCP	Email receiving
IMAP	143	TCP	Email sync
SNMP	161	UDP	Network management
RDP	3389	TCP	Remote desktop

OSI Model — Quick Recall

Layer	Name	Mnemonic	Key Devices	Protocols
7	Application	All	—	HTTP, DNS, SMTP
6	Presentation	People	—	SSL, JPEG
5	Session	Seem	—	NetBIOS
4	Transport	To	—	TCP, UDP
3	Network	Need	Router	IP, ICMP
2	Data Link	Data	Switch, Bridge	Ethernet
1	Physical	Processing	Hub, NIC	Cables

Common Interview Questions Summary

- ? **Top Q1:** OSI ke 7 layers explain karo — name + function.
- ? **Top Q2:** TCP 3-way handshake explain karo.
- ? **Top Q3:** Jab browser mein URL type karte ho toh kya hota hai?
- ? **Top Q4:** TCP vs UDP — differences + use cases.
- ? **Top Q5:** Subnetting kya hai? 192.168.1.0/25 mein kitne hosts?

- ? **Top Q6:** HTTP vs HTTPS kya difference hai?
- ? **Top Q7:** DNS kaise kaam karta hai?
- ? **Top Q8:** Router vs Switch vs Hub mein kya difference hai?
- ? **Top Q9:** DDoS attack kya hai? Kaise prevent karte hain?
- ? **Top Q10:** NAT kya hai aur kyun use karte hain?